AI Report

We predict this text is

Human Generated

AI Probability

31%

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Tugas Sistem Pemodelan Muhammad Shawfan Zaky.pdf - 10/22/2026

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| Tugas | : | Mencari Satu Kasus Yang Berkaitan Dengan Ketidakpastian, Probabilitas Tidak Boleh Sama, Mengumpulkan Variabel, Operasi Dan Hasilnya.
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Study Case

Simulasi Ketidakpastian Tingkat Kualitas Udara Harian (PM2.5 / ISPU) Menggunakan Bilangan Acak (Metode Monte Carlo)

Kualitas udara dan konsentrasi polutan \(PM{2.5} \) di wilayah perkotaan setiap harinya tidak selalu sama karena dipengaruhi oleh kepadatan lalu lintas, aktivitas industri, kecepatan angin, dan curah hujan.
Karena tingkat pencemaran udara di masa mendatang tidak dapat diketahui secara pasti, saya membuat simulasi menggunakan distribusi probabilitas dan bilangan acak untuk memperkirakan indeks kualitas udara harian berdasarkan data historis (airquality_dataset.csv).

Tabel Variabel Simulasi:

| Variabel | Keterangan | Nilai |

| --- | --- | --- |

| X | Nilai konsentrasi \(PM_{2.5} \) / ISPU per hari | Belum diketahui |

| P(X) | Probabilitas kategori kualitas udara | 0, 20; 0, 50; 0, 30 |

| R | Bilangan acak | 0 sampai kurang dari 1 |

| D | Jumlah hari simulasi | 10 hari |

| T | Total akumulasi nilai indeks selama simulasi | Hasil penjumlahan |

Kumpulan Variabel

Berdasarkan data pengamatan historis, saya menetapkan tiga kategori kemungkinan pada tingkat kualitas udara harian dengan probabilitas yang berbeda:

| Kategori Kualitas Udara | Rentang Nilai (PM2.5 / ISPU) | Nilai Tengah (X) | Probabilitas |

| --- | --- | --- | --- |

| Baik | 30 – 50 | 40 | 20% (0, 20) |

| Sedang | 51 – 99 | 75 | 50% (0, 50) |

| Tidak Sehat | 100 – 140 | 120 | 30% (0, 30) |

| Total | | | 100% (1, 00) |

Probabilitas ini ditetapkan berdasarkan distribusi frekuensi kemunculan kondisi kualitas udara agar simulasi Monte Carlo dapat dilakukan.

Langkah 1: Membentuk interval probabilitas kumulatif

Karena bilangan acak R berada pada rentang 0 sampai kurang dari 1:

\(0 \leq R < 0, 20 \rightarrow\) Kategori Baik, pilih nilai tengah 40 sebagai nilai simulasi.

\(0,20 \leq R < 0,70 \rightarrow\) Kategori Sedang, pilih nilai tengah 75 sebagai nilai simulasi.

\(0,70 \leq R < 1,00 \rightarrow\) Kategori Tidak Sehat, pilih nilai tengah 120 sebagai nilai simulasi.

Langkah 2: Menghasilkan bilangan acak

Menggunakan fungsi pembangkit bilangan acak kontinu U(0, 1) pada Python (np.random.uniform(0, 1)) atau fungsi =RAND() di Excel untuk menghasilkan satu bilangan acak R setiap harinya.

Langkah 3: Menghitung hasil operasi

Menentukan nilai indeks kualitas udara (X_i) setiap hari berdasarkan interval bilangan acak, kemudian menjumlahkan seluruh nilai selama 10 hari dan menghitung rata-rata hariannya dengan rumus sebagai berikut:

\text{Total Akumulasi Indeks: } T = \sum{i=1}^{10} Xi

\text{Rata-rata Kualitas Udara Harian: } \overline{X} = \frac{T}{10}

Perhitungan Totalnya

| Hari ke- | Bilangan Acak (R) | Kategori Kualitas Udara | Nilai Simulasi (X) |

| --- | --- | --- | --- |

| 1 | 0,14 | Baik | 40 |

| 2 | 0,48 | Sedang | 75 |

| 3 | 0,83 | Tidak Sehat | 120 |

| 4 | 0,32 | Sedang | 75 |

| 5 | 0,65 | Sedang | 75 |

| 6 | 0,09 | Baik | 40 |

| 7 | 0,91 | Tidak Sehat | 120 |

| 8 | 0,54 | Sedang | 75 |

| 9 | 0,76 | Tidak Sehat | 120 |

| 10 | 0,27 | Sedang | 75 |

| Total (T) | | | 815 |

Perhitungan Rata-rata Harian:

$\overline{X} = \frac{815}{10} = 81,5$

Kesimpulan:

Artinya, berdasarkan hasil simulasi selama 10 hari pengamatan menggunakan bilangan acak, diperoleh total akumulasi indeks sebesar 815 dengan rata-rata tingkat kualitas udara harian sebesar 81,5 (berada pada kategori Sedang), serta terdapat 3 hari yang diprediksi mengalami kondisi udara Tidak Sehat.

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High Human Impact High AI Impact

FAQs

What is GPTZero?

GPTZero is the leading AI detector for checking whether a document was written by a large language model such as ChatGPT. GPTZero detects AI on sentence, paragraph, and document level. Our model was trained on a large, diverse corpus of human-written and AI-generated text with support for English, Spanish, French, German, and other languages. To date, GPTZero has served over 20 million users around the world, and works with over 100 organizations in education, hiring, publishing, legal, and more.

When should I use GPTZero?

Our users have seen the use of AI-generated text proliferate into education, certification, hiring and recruitment, social writing platforms, disinformation, and beyond. We've created GPTZero as a tool to highlight the possible use of AI in writing text. In particular, we focus on classifying AI use in prose. Overall, our classifier is intended to be used to flag situations in which a conversation can be started (for example, between educators and students) to drive further inquiry and spread awareness of the risks of using AI in written work.

Does GPTZero only detect ChatGPT outputs?

No, GPTZero works robustly across a range of AI language models, including but not limited to ChatGPT, GPT-5, GPT-4, GPT-3, Gemini, Claude, and AI services based on those models.

What are the limitations of the classifier?

The nature of AI-generated content is changing constantly. As such, these results should not be used to punish students. We recommend educators to use our behind-the-scene Writing Reports as part of a holistic assessment of student work. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Instead, we recommend educators take approaches that give students the opportunity to demonstrate their understanding in a controlled environment and craft assignments that cannot be solved with AI. Our classifier is not trained to identify AI-generated text after it has been heavily modified after generation (although we estimate this is a minority of the uses for AI-generation at the moment). Currently, our classifier can sometimes flag other machine-generated or highly procedural text as AI-generated, and as such, should be used on more descriptive portions of text.

I'm an educator who has found AI-generated text by my students. What do I do?

Firstly, at GPTZero, we don't believe that any AI detector is perfect. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Nonetheless, we recommend that educators can do the following when they get a positive detection: Ask students to demonstrate their understanding in a controlled environment, whether that is through an in-person assessment, or through an editor that can track their edit history (for instance, using our Writing Reports through Google Docs). Check out our list of several recommendations on types of assignments that are difficult to solve with AI.

Ask the student if they can produce artifacts of their writing process, whether it is drafts, revision histories, or brainstorming notes. For example, if the editor they used to write the text has an edit history (such as Google Docs), and it was typed out with several edits over a reasonable period of time, it is likely the student work is authentic. You can use GPTZero's Writing Reports to replay the student's writing process, and view signals that indicate the authenticity of the work.

See if there is a history of AI-generated text in the student's work. We recommend looking for a long-term pattern of AI use, as opposed to a single instance, in order to determine whether the student is using AI.